

Chen-Yu Kuo 郭宸毓

Systems and full-stack engineer — real-time earthquake early warning, from sensor firmware to the app that rings

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I founded and lead **ExpTech**, a 29-person volunteer engineering organisation that builds and operates Taiwan's largest independent earthquake early-warning platform. I work the whole path a seismic alert travels: the ESP32 accelerometer firmware I wrote register by register, the multi-region ingest and alerting fleet it reports to, the ground-motion models that decide what the alert says, and the Flutter app that over 100,000 people installed to receive it.

EXPERIENCE

Founder & Lead Engineer · ExpTech (探索科技)

Oct 2020 — Present

Taiwan — volunteer engineering organisation, 29 members, 234 repositories

- Lead the organisation as one of its two administrators. Authored **13,411** commits inside it (**19,284** across GitHub, **477** pull requests) and am the top contributor on every flagship codebase: DPIP **838/2,339**, TREM-Lite **448/847**, platform API **480/803** — and sole author of the 665-commit production infrastructure repository.
- Signed a data agreement with Taiwan's **Central Weather Administration (CWA)** in late 2022; reported by 親子天下 as the Seismological Center's first non-corporate partner. Invited back to the centre 10+ times as a collaborator (天下雜誌, Feb 2026).
- Ship under real stakes to real users: **100,000+** Google Play installs and **4.2/5** from 1,378 App Store ratings for DPIP; **328,542** installer downloads across TREM desktop releases. During the M7.2 Hualien earthquake (3 Apr 2024) the network recorded 800+ events in a single day.
- Grew the codebase from a school-project origin into a fleet spanning firmware, 37 Go services, Flutter clients and ML pipelines — recruiting and reviewing a distributed team of student engineers.

SELECTED ENGINEERING WORK

Ground-motion & phase-picking models Python · PyTorch · SeisBench · ONNX · sole author huggingface.co/YuYu1015

- Refit Taiwan's shaking-intensity model on **45,398** CWA PGA/PGV/intensity observations (2000–2025), replacing a legacy equation that over-amplified with magnitude. Added a direct PGV ground-motion prediction equation and corrected a site-amplification double-count that inflated estimates roughly 2×. Held-out result: **62.0% exact intensity vs 39.7%** for the model then in production, MAE **0.400 vs 0.671**, with 0% false alarms. Released as *IntensityGMM-TW-45k-v1*.
- Trained a 50 Hz PhaseNet P/S phase picker — a 6,457-line pipeline covering dataset ingest, noise injection, evaluation and ONNX export. Diagnosed a windowing bug in which the source window was shorter than the model window, teaching the model that P always sits mid-window and emitting a false pick every 45 s stride. Released as *PhaseNet-50Hz-268k-v1*.
- Built a P-window → S-amplitude regressor: 1–5 s post-P features (τ_c after Wu & Kanamori 2005, spectral centroid and bandwidth, kurtosis, zero crossings) into CNN and tabular models.

ES-Net — seismometer firmware & national sensor network C++ · ESP32 · FreeRTOS · sole author, 7,759 LOC

- Hand-wrote an **LSM6DS3 accelerometer driver at register level** — own register map, WHO_AM_I, CTRL1_XL, INT1_CTRL and FIFO paths — rather than depending on a vendor library.
- On-device **JMA seismic-intensity** computation: an IIR filter bank plus a binary indexed tree for the running amplitude percentile the JMA scale requires.
- Timing is the hard problem in a distributed seismometer: SNTP using SMOOTH sync mode so the clock slews rather than steps mid-waveform, on a 60 s interval — alongside hardware and software watchdogs, OTA update, and a documented custom binary wire protocol with TOTP station authentication.
- Feeds **200+ TREM-Net stations** nationwide; the first-generation units were self-assembled for under NT\$800 each.

TREM — real-time earthquake monitoring desktop platform JavaScript/Electron · Rust · 93 stars · top contributor 448/847 ExptechTW/TREM-Lite

- Desktop client that renders live ground motion from TREM-Net and CWA feeds. **328,542 installer downloads** across 74 releases of the current and previous generations, shipped continuously since Jan 2023.
- Designed the plugin system third parties extend the client through and maintain its registry (417 commits); wrote the companion trem-monitor station-health service in Rust (105/150 commits).

QZSS L1S disaster-message decoder C++ · sole author · ~12k LOC with unit tests

- Decodes QZSS L1S 250-bit SBAS frames end to end — preamble, 6-bit message type, 212-bit payload and CRC-24Q — including type 43 **DC Report** (JMA disaster broadcasts, IS-QZSS-DCR) and type 44 DCX.

Production infrastructure Go · eBPF/XDP · OpenResty · CoreDNS · sole author

- kekkaï** — packet filter in Go using cilium/ebpf, with an XDP ingress program and a TC egress hook;

LRU_HASH/ARRAY/RINGBUF maps drive per-IP rate limiting, allow/blocklists and port policy, with an emergency detach path.

- **Nginx/OpenResty fleet** — 23 upstreams across nodes in Taipei, Kaohsiung, Tainan, Tokyo and Osaka, with Lua modules for traffic accounting, NTP and image proxying.
- **Split-horizon CoreDNS** with a custom Go dnstats plugin (miekg/dns) exposing per-domain and per-client query statistics plus Prometheus metrics.
- Sole author too of the platform's NTP server, API gateway and Meshtastic LoRa ingest service — among the 37 Go services the organisation runs.

DPIP — cross-platform disaster alerting Dart/Flutter · TypeScript · top contributor [ExpTechTW/DPIP](#)

- iOS and Android client integrating TREM-Net and CWA feeds with push alerting, MapLibre layers and 10 interface languages. Wrote the repository's test suite and its custom commit-gate tooling outright, and authored the backend service dpip-server-ts in full.

LLM inference & model compression PyTorch · vLLM · llm-compressor · 27 models, ~10,600 downloads

- Quantise large open models to **NVFP4** (TensorRT Model Optimizer), INT4-AutoRound and GGUF for vLLM and llama.cpp serving, including MoE architectures up to 35B.
- Implemented **refusal-direction ablation** — deriving the refusal direction from paired prompts and orthogonalising it out of the weight matrices — extending the upstream reference implementation, and released the *Ornith-1.0* 9B/35B line including a DPO-tuned variant.

Security & incident response honeypot · malware triage

- Operated a honeypot that captured **in-the-wild exploitation of CVE-2025-55182** (React Server Components, CVSS 10.0) within days of its December 2025 disclosure; documented six malware samples with verified hashes and fifteen attacker source addresses.

TECHNICAL SKILLS

Languages	TypeScript/JavaScript, Go, Python, Dart, C/C++, Rust, Lua, Shell
Systems	Linux, eBPF/XDP, Docker, Nginx/OpenResty, CoreDNS, WebSocket & SSE, NTP, Prometheus, CI/CD
Embedded	ESP32/Arduino, FreeRTOS, I ² C/SPI, LSM6DS3, GNSS (QZSS L1S), LoRa/Meshtastic, RTL-SDR
ML	PyTorch, SeisBench, ONNX, scikit-learn, vLLM, llm-compressor, TensorRT Model Optimizer
Domain	Earthquake early warning, GMPE calibration, CWA/JMA intensity scales, phase picking, NonLinLoc travel-time grids

EDUCATION

B.S., Electrical Engineering · National Kaohsiung University of Science and Technology Kaohsiung, Taiwan (NKUST)

OPEN SOURCE & COMMUNITY

- **12 merged pull requests** to projects outside my own organisation — a rendering-performance fix to **AnyShake's** spectrogram-js (an open-source seismograph suite), a Leaflet → MapLibre migration for `smc.peering.tw`, and Flutter toolchain upgrades to **NKUST-AP-Flutter**, my university's official student app.

PRESS & RECOGNITION

- 天下雜誌 (CommonWealth Magazine), Feb 2026 — quoted by name as an ExpTech collaborator in a profile of the CWA Seismological Center.
- 親子天下翻轉教育, Jul 2024 — feature on the DPIP team; the app reached roughly 500,000 downloads following the 3 April 2024 earthquake.

Provenance — commit counts are from GitHub's commit-search index and per-repository contributor statistics (retrieved Sep 2026). The TREM download figure counts installer assets only (.exe/.dmg/.deb/.AppImage/.zip), excluding 4.3M auto-updater metadata requests that GitHub also reports as downloads.